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Maritime Autonomous Surface Ships



This white paper was researched and authored by **Chanda Wagh**, Technical Writer & Quality Coordinator with hands-on experience documenting AI-based Unmanned Surface Vessel (USV) systems, collision avoidance platforms, and defence-grade compliance documentation.

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1 Executive Summary

1.1 Mission-Critical Capabilities

A modular, hardware-agnostic software suite delivering adaptive maritime surveillance and AI-driven collision avoidance for USVs. Designed for seamless integration with both legacy and next-generation platforms, it combines advanced perception, autonomous navigation, and real-time connectivity to ensure reliable performance in complex and contested maritime environments. Its scalable architecture supports seamless capability upgrades as mission demands evolve, ensuring long-term operational viability

1.2 Strategic Advantage

The USV removes personnel risk while enabling persistent maritime surveillance and mission support across defense, commercial, environmental, and emergency-response domains. It can deliver actionable intelligence through fully autonomous operations in demanding sea states, using adaptive path-planning algorithms that respond to evolving threats, operational demands, and dynamic environmental conditions



2 Integrated Sensor Architecture

Maritime Radar Systems

X-band and S-band radar arrays can deliver full 360° situational coverage for vessel detection, collision avoidance, and navigation support, continuously tracking targets and providing automated threat classification for enhanced maritime awareness.

Electro-Optical Suite

The software can ingest and process multi-spectral data from dual thermal and visual camera systems on gyro-stabilized gimbals, ensuring high-fidelity imaging in all lighting and weather. It can interpret long-wave infrared streams for heat-signature detection and analyze RGB feeds to support precise target identification, classification, and verification.

Navigation Sensors

Military-grade IMU and INS providing positional accuracy within 1 metre. RTK-GPS with GNSS redundancy ensures continuous navigation even in GPS-denied environments through sensor fusion algorithms.

Environmental Monitoring

Integrated depth sounders, anemometers, and temperature sensors continuously assess operating conditions. Their real-time environmental data can flow directly into the software's autonomous decision-making algorithms, enabling optimized route planning and safer navigation.

3 AI-Driven Autonomous Navigation



Situational Awareness

Machine learning models fuse radar tracks, AIS data, and visual detections to classify contacts, predict intentions, and assess collision risk in real-time.



Dynamic Path Planning

AI algorithms compute collision-free trajectories compliant with International Regulations for Preventing Collisions at Sea (COLREGs), adapting to traffic density and mission priorities.



Threat Response

Autonomous decision-making enables rapid response to hostile contacts, environmental hazards, or system failures, executing pre-programmed tactical behaviours without operator intervention.

4 Ground Station Command & Control



Real-Time Monitoring

The software supports high-bandwidth satellite communication, enabling continuous streaming of sensor data, vessel telemetry, and environmental parameters to shore-based operators with minimal end-to-end latency.



Remote Intervention

Operators can take direct control or adjust mission parameters at any time through the command interface. Built-in emergency-stop protocols allow instant intervention in response to unexpected conditions.



Mission Planning

The intuitive interface enables rapid mission planning with automated route generation, power-use estimates, and risk assessments, providing operators with clear validation before execution.



5 Future Development Roadmap

1

2026: Enhanced Autonomy

Future USV systems will implement swarm intelligence to enable coordinated, multi-vessel operations. Advanced AI algorithms will incorporate predictive threat assessment and collaborative mission planning, allowing USV fleets to operate autonomously and synergistically across complex maritime environments

2

2027: Expanded Capabilities

Integration of underwater drone deployment systems for subsurface surveying. Advanced electronic warfare suites providing signals intelligence gathering and communication jamming capabilities.

3

2028: Global Deployment

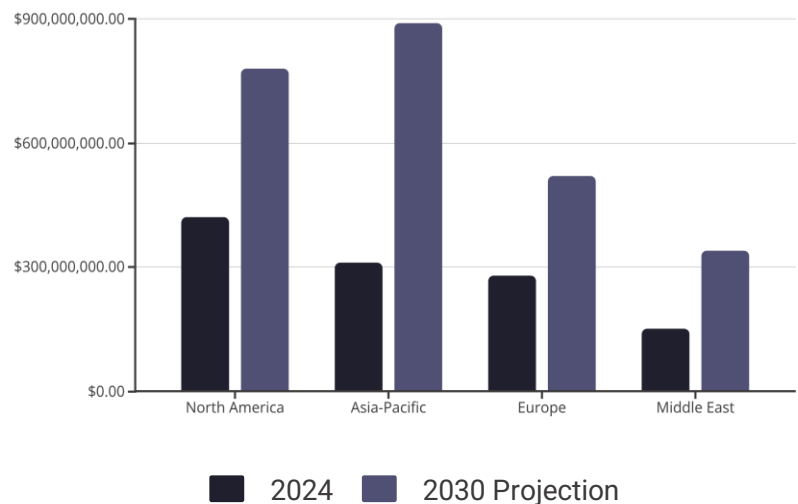
Future USV platforms are expected to include Arctic-hardened variants capable of extended polar and extreme-weather operations. Evolving modular mission bays will support even faster reconfiguration, enabling next-generation roles in autonomous hydrographic surveying, long-endurance maritime patrol, precision interdiction, and integrated multi-domain mission support

4

2029: AI Maturation

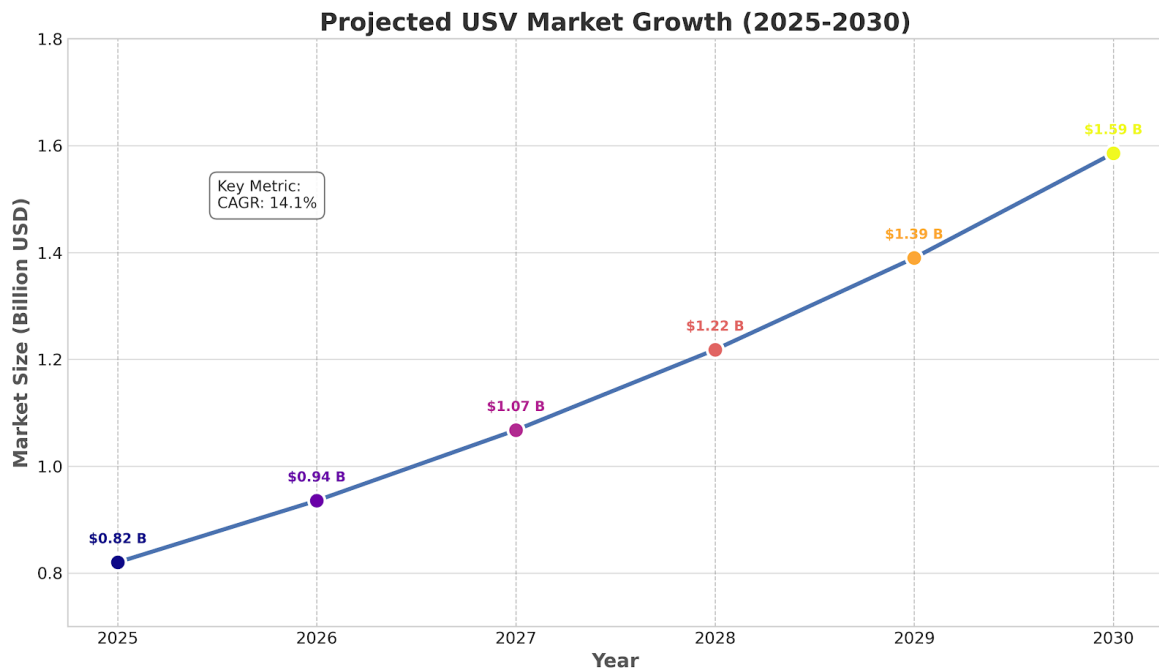
Fully autonomous decision-making for complex tactical scenarios. Reduced ground station dependency enabling operations in communications-denied environments with pre-programmed response protocols.

6 Market Drivers & Growth Projections



The global military USV market demonstrates robust expansion, driven by strategic imperatives to enhance maritime domain awareness whilst reducing personnel exposure to hostile environments.

7 Market Drivers & Growth Projections



8 Case Study

- TideWise (Brazil): Deployed EIVA's NaviSuite Kuda for hydrographic surveys, achieving high-quality data collection and operational flexibility with modular software.
- Sonardyne Navigation: Utilized advanced navigation and sensor fusion in long-duration survey missions under challenging sea conditions, demonstrating high reliability and data accuracy.
- SEA-KIT: Employed Cambridge Pixel's radar and situational awareness software to enhance remote operations and obstacle detection during offshore surveys.
- AI Collision Avoidance: Implemented real-time, AI-driven algorithms that optimize autonomous maneuvering and collision avoidance, significantly improving safety in complex, dynamic environments

9 Conclusion

USVs are rapidly becoming essential assets for modern maritime missions, delivering persistent coverage, reduced human risk, and high-precision autonomy. Advances in sensors, AI navigation, and modular architectures now enable reliable operation in complex and contested environments. As capabilities evolve toward swarm coordination and predictive decision-making, USVs will further expand mission effectiveness across defense, commercial, and environmental domains. Their growing role signals a clear shift towards smarter, safer, and more efficient maritime operations.

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